EMBL's European Bioinformatics Institute

Unleashing the potential of big data in biology

Dr Colman O'Cathail
ENA Project Lead and Product Owner

EMBL-EBI

What is EMBL-EBI?

- World leading source of public biomolecular data
- Our vision is to benefit humankind by advancing scientific discovery and impact through bioinformatics.
- Part of the European Molecular Biology Laboratory (EMBL), Europe's flagship laboratory for the life sciences.



EMBL-EBI

Our mission



Deliver data resources	Perform excellent research	Train the next generation of scientists	Engage with industry	Coordinate bioinformatics in Europe

EMBL-EBI

EMBL-EBI in 2025

646
established members of personnel

121 million requests to our data resources on an average day



from **127 million** unique IP addresses annually*

>123,200
scientific publications enabled by EMBL-EBI data resources

259
journal papers published

430,223
users accessed our online training

1,739



attendees to our Industry Programme workshops

1,048



participants to our public engagement events

*EMBL-EBI has seen a large increase in IP addresses in 2025, reflecting the increased use of AI-powered automated requests

EMBL-EBI

The European Molecular Biology Laboratory



EMBL-EBI

Bioinformatics



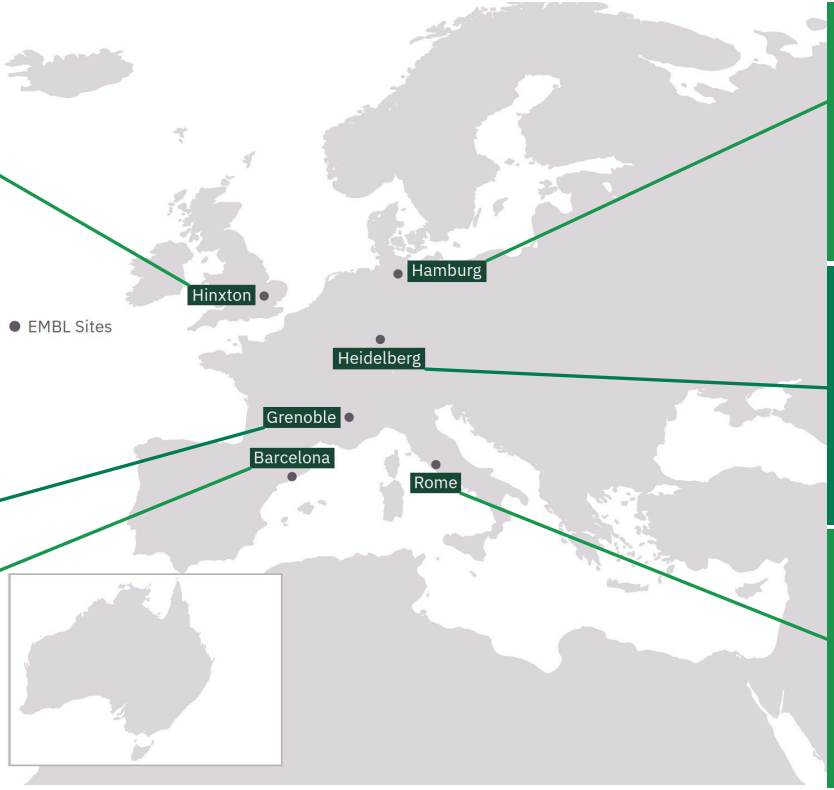
Grenoble

Structural biology



Barcelona

Tissue biology and disease modelling



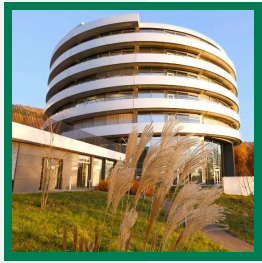
Hamburg

Structural biology



Heidelberg

Life sciences



Rome

Epigenetics and neurobiology



EMBL-EBI

EMBL member states

Member states (29)

Austria 1974
Denmark 1974
France 1974
Germany 1974
Israel 1974
Italy 1974
Netherlands 1974
Sweden 1974
Switzerland 1974
United Kingdom 1974
Finland 1984
Greece 1984
Norway 1985
Spain 1986
Belgium 1990
Portugal 1998
Ireland 2003
Iceland 2005
Croatia 2006
Luxembourg 2007
Czech Republic 2014
Malta 2016
Hungary 2017
Slovakia 2018
Montenegro 2018
Lithuania 2019
Poland 2019
Estonia 2023
Latvia 2023



Associate member state

Australia 2008



Prospect member states

Bulgaria



Serbia



Sources of funding

- EMBL-EBI is primarily funded by EMBL's member states.
- Other major funders:
  - European Commission
  - UK Research and Innovation
  - National Institutes of Health
  - Wellcome
  - Industry Programme

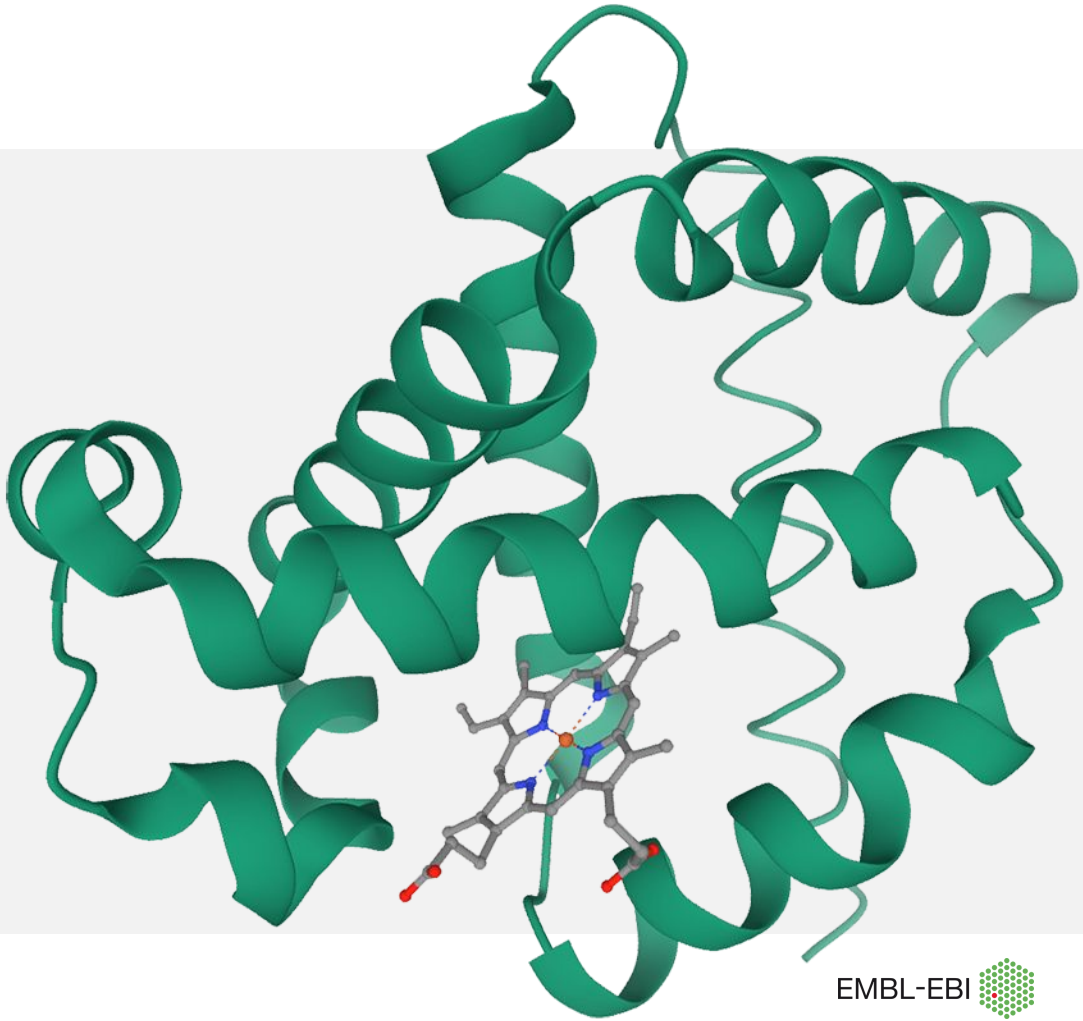






Bioinformatics

The science of analysing, managing, storing and sharing biological information



EMBL-EBI

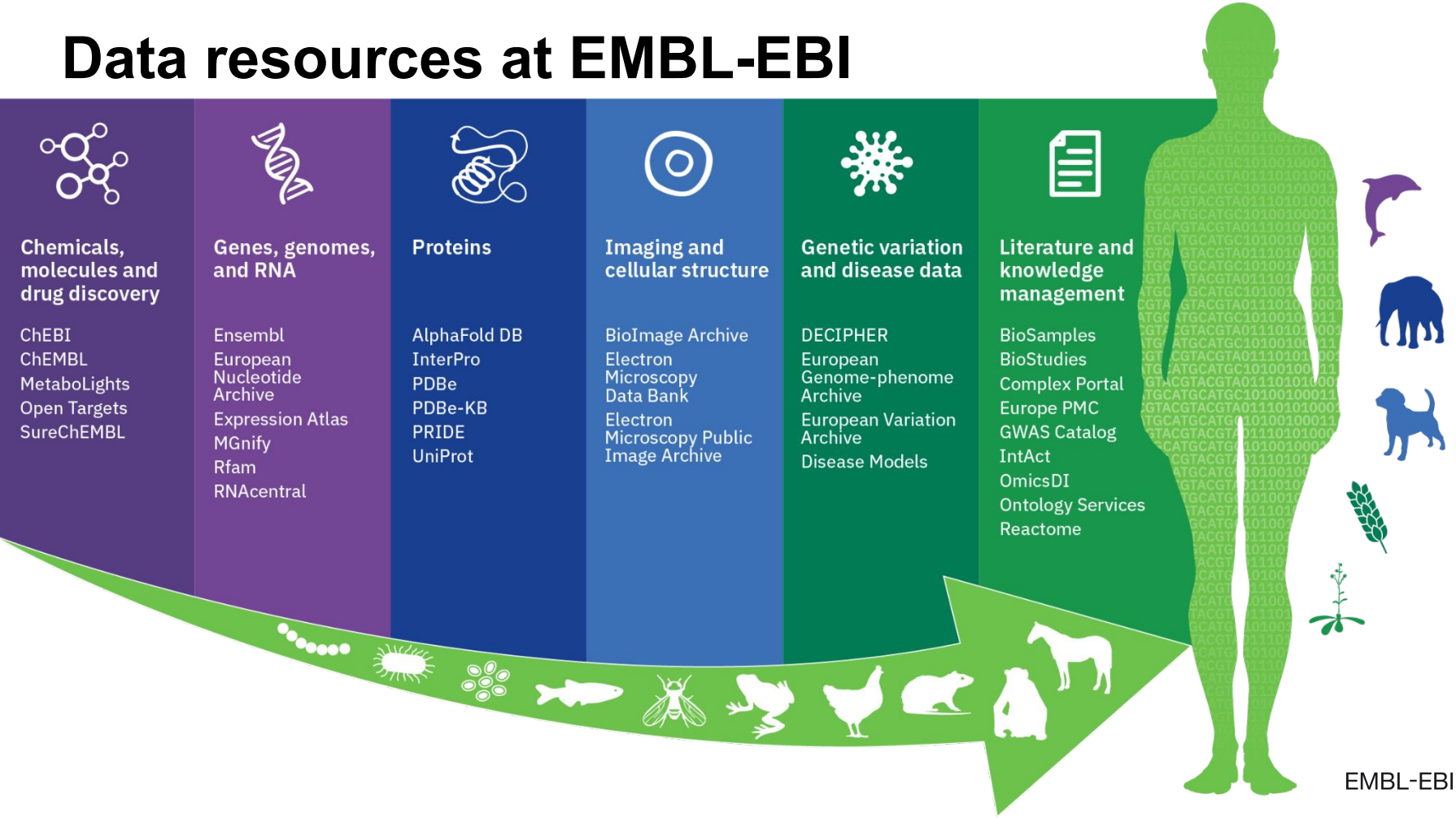
Data resources to support life science research



www.ebi.ac.uk/**services**

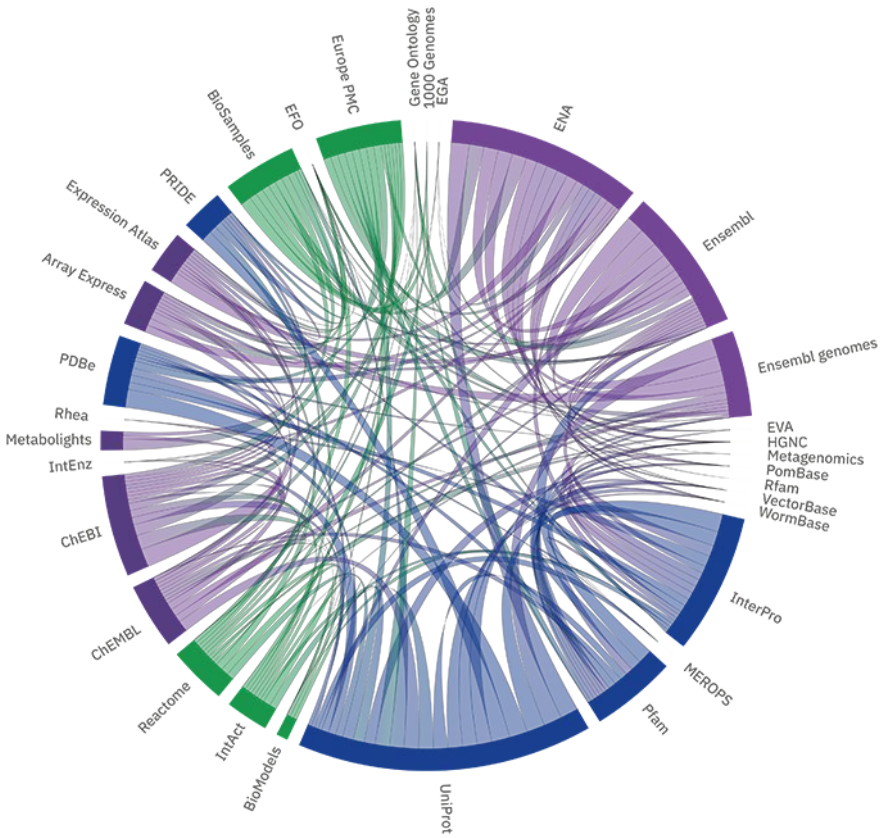
EMBL-EBI

Data resources at EMBL-EBI



Database interactions

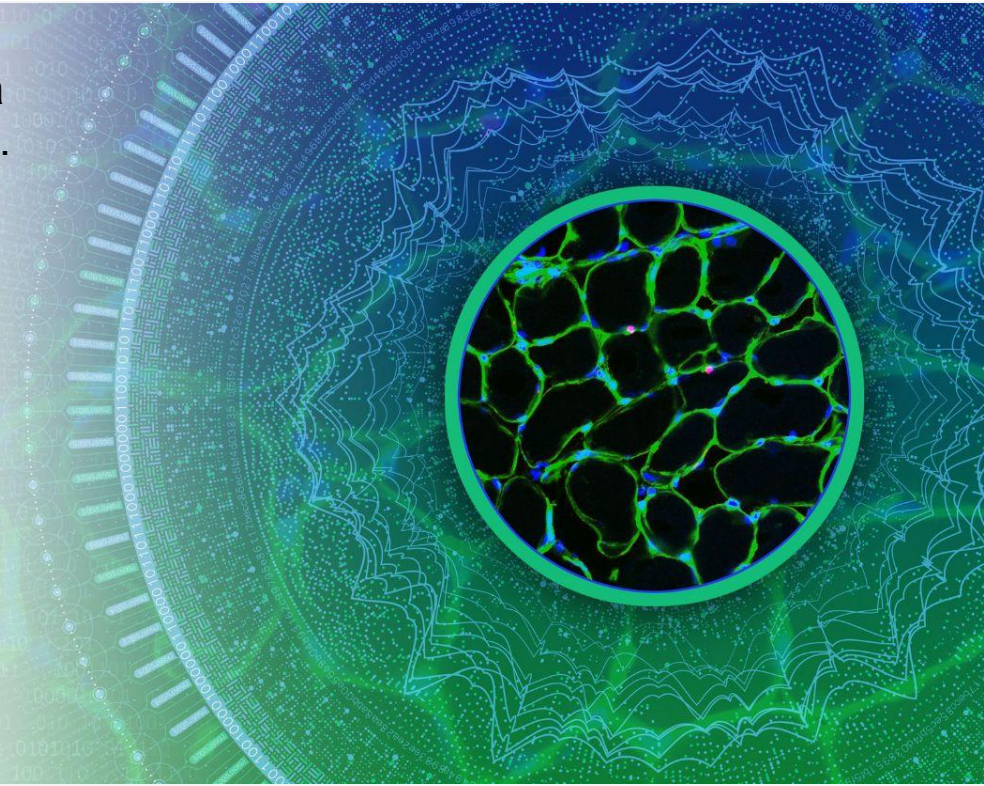
- Our collaborative community facilitates social, scientific and technical interactions
- This image shows internal interactions between data resources, as determined by the exchange of data.
- The width of each internal arc is weighted according to the number of different data types exchanged.



EMBL-EBI

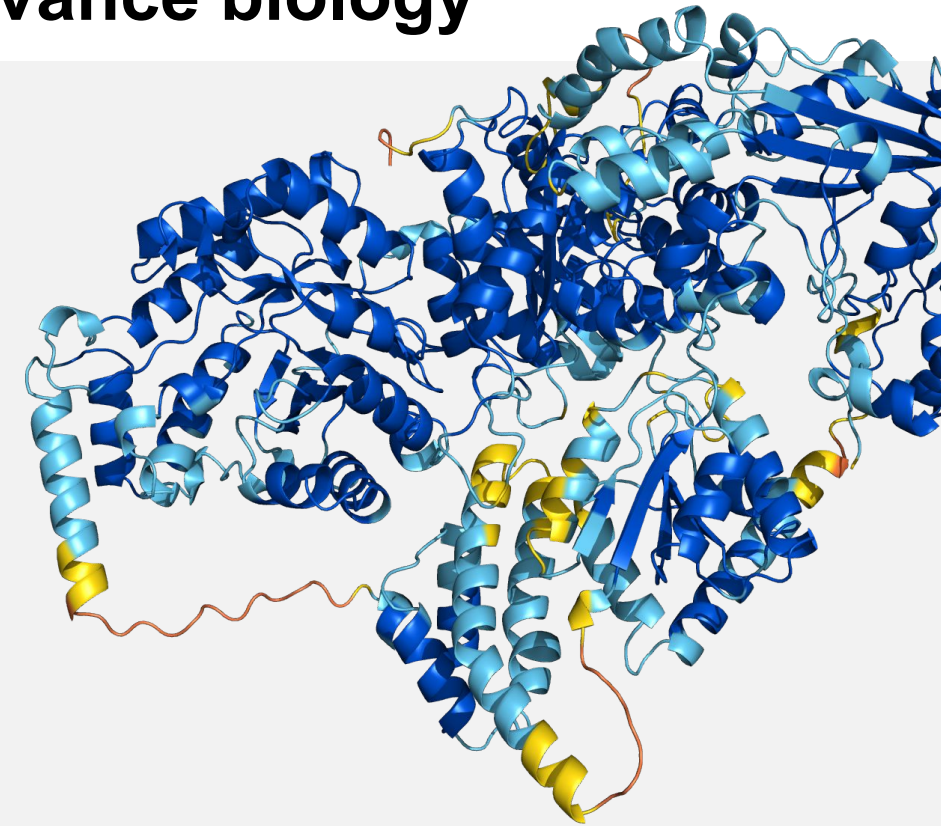
Artificial intelligence and machine learning

- Researchers and data scientists who develop AI algorithms use high quality data managed by EMBL-EBI to train AI systems.
- We add value to data submitted to our resources through expert curation and annotation.
- When appropriate, we integrate AI predictions into the bioinformatics ecosystem.



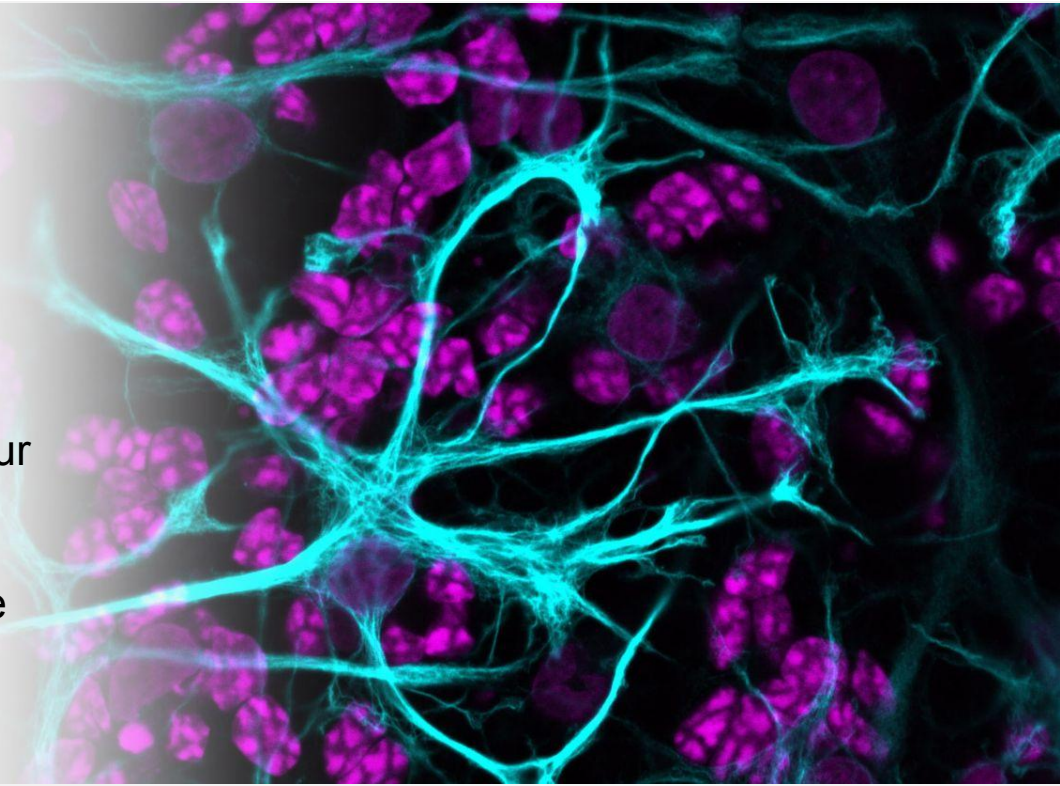
AlphaFold – using AI to advance biology

- Google DeepMind used public databases UniProt, PDB and MGnify to train the AlphaFold algorithm, which predicted the structure of all known proteins.
- EMBL-EBI made +200 million protein and complex structure predictions openly available in the AlphaFold Database.
- Potential applications of AlphaFold:
  - Drug discovery and vaccine development
  - Neglected diseases
  - Plastic pollution



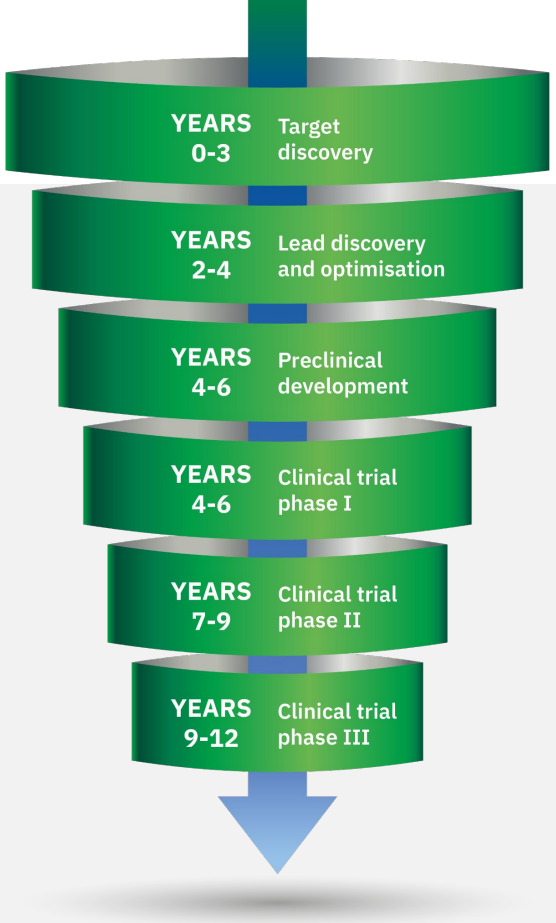
Biological imaging

- Light and electron microscopy are among the fastest growing types of biological big data.
- EMBL-EBI works with imaging user communities to create common standards and establish foundational resources.
- Researchers and data scientists use our resources, such as BioImage Archive and Electron Microscopy Databank, to train AI algorithms, understand disease and develop new drugs.



Drug discovery

- Developing a new drug can take over ten years and cost millions.
- Pharmaceutical companies use bioinformatics and AI to speed up the process and increase chances of success.
- EMBL-EBI data resources, including ChEMBL, PDBe, AlphaFold DB, Reactome and Open Targets support all stages of drug discovery



Genomic medicine

- Healthcare experts, including NHS England and the German Genome Phenome Archive, use EMBL-EBI data resources to deliver thousands of genetic diagnostics to patients every month.
- We develop analysis tools to understand genome sequencing data from rare disease and cancer patients, to improve diagnosis and treatment.
- We help train the next generation of genomic medicine experts.



Biosecurity

- Healthcare and public health specialists use EMBL-EBI's Pathogens Portal to access data about 200,000 pathogen species and strains
- We support countries to build national Pathogen Portals, to get insights from patient data in their healthcare systems (e.g Swedish Pathogen Portal)
- This helps to monitor outbreaks and antimicrobial resistance, identify new pathogen strains, and develop new treatments and vaccines.



Biodiversity conservation

- Ecologists and conservation biologists use bioinformatics and genomics to:
  - understand and protect biodiversity
  - develop clean technologies to reduce environmental pollution



TREC

- EMBL's scientific expedition to study European coastlines from molecules to ecosystems.
- The data gathered will be made publicly-available through EMBL-EBI resources.

TREC
Traversing European Coastlines

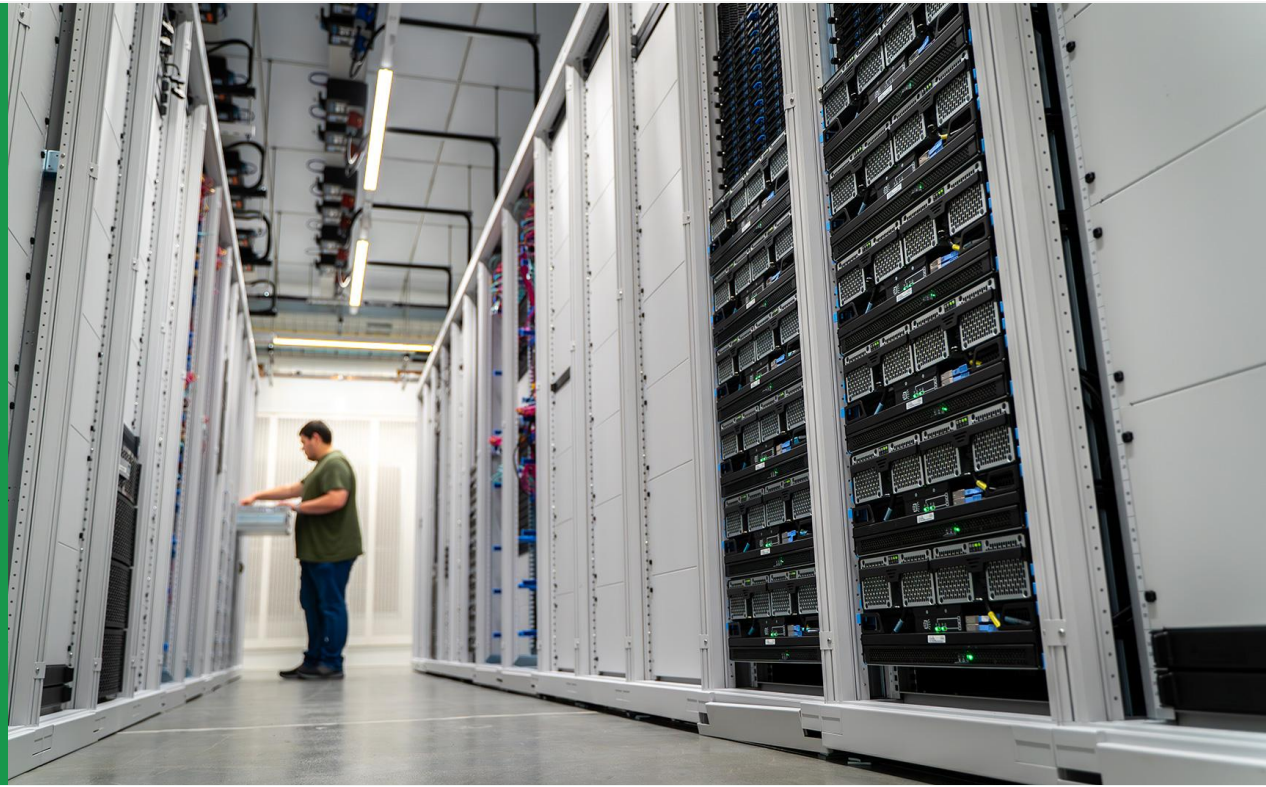
Sustainable food production

- Farmers, animal breeders and agricultural scientists use our data resources to:
  - Identify which species and cultivars are tolerant to climate change while still providing optimum nutrition
  - Inform strategies for dealing with pests while protecting pollinators
  - Link genes to traits to make food production more robust, effective and sustainable



World-leading technical infrastructure

- Underpins our open data resources
- Facilitates easy access to data for millions of scientists
- Robust and flexible technology allows us to deal with data on a massive scale



Technical services

Web services
(web production, web dev)

Virtualisation & cloud
(virtual machines, cloud storage and technology)

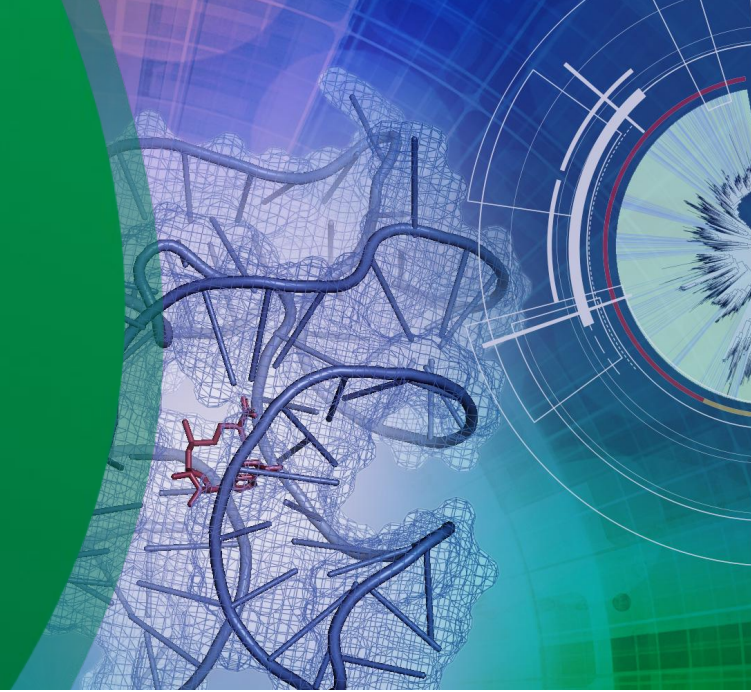
Data storage
(data centres, databases, networking)

Software development
(FIRE, DevOps)

Collaborations
(European Open Science Cloud, Archiver, Bioexcel etc)

EMBL-EBI

Research



www.ebi.ac.uk/**research**

EMBL-EBI

Data-driven discovery

Based on the Wellcome Genome Campus, we are in a unique setting for research.

- Rich technical environment
- One of the world's highest concentrations of expertise in genomics and computational biology
- Research and data resources are mutually supportive



Research groups at EMBL-EBI


Alex Bateman


Ewan Birney


Tim Coorens


Isidro Cortés-Ciriano


Jess Ewald


Rob Finn


Nick Goldman


John Lees


Evangelia Petsalaki


Julio Saez-Rodriguez

EMBL-EBI

The EMBL PhD Programme

First-class training and education

- International
- Dedicated mentoring
- Early independence in research
- Europe's 'best in class' PhD programme for the life sciences
- EMBL-EBI graduates receive University of Cambridge degree



EMBL-EBI

Postdoctoral opportunities at EMBL-EBI



Ongoing recruitment

Our research groups and service teams advertise funded postdoctoral opportunities all year round.

EIPOD
exploring life in context

Interdisciplinary

EIPOD projects bring together scientific fields that are usually separate, or transfer techniques to a novel context.

ESPOD
EMBL-EBI/Sanger postdoctoral fellowship

Genome campus

ESPODs: joint Sanger–EMBL-EBI fellowships combine experimental and computational approaches.

EMBL-EBI

Bioinformatics training



www.ebi.ac.uk/**training**

EMBL-EBI

Vision

A programme that enables scientists at all levels, regardless of location, sector or career stage, to make full use of public biodata and to share their expertise openly with others



EMBL-EBI Training in a nutshell



Live training

Join us – virtually or in person – to learn with other scientists from around the world



On-demand training

Dive straight into learning through our online tutorials, course materials and recorded webinars



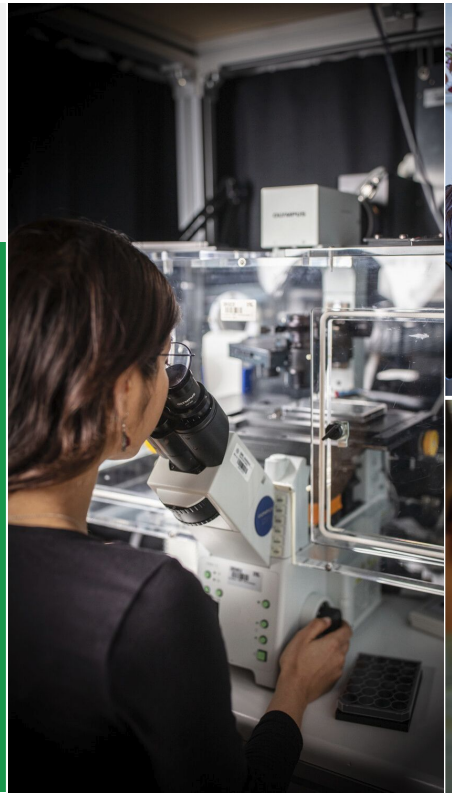
Support for trainers

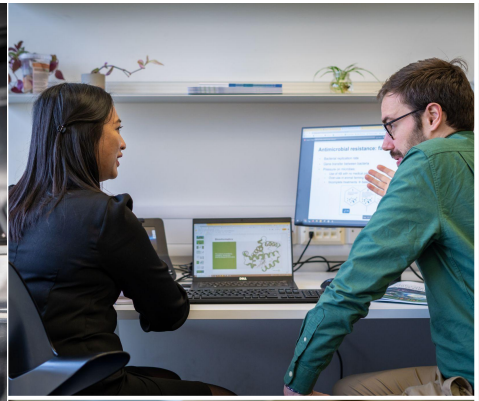
We support you and your courses with open resources and skills development

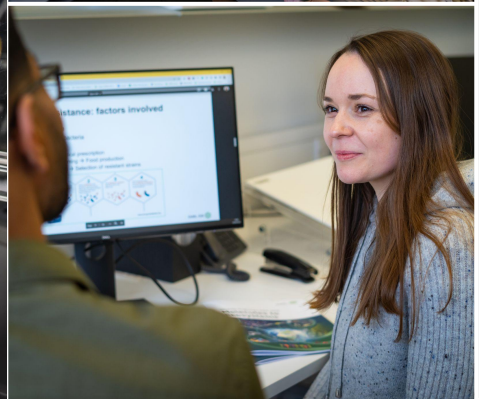
ARISE: Shaping the next generation of infrastructure leaders

ARISE

ARISE is a fellowship programme for research infrastructure scientists who are trained to both **advance technology and methods in life sciences** and to **manage service-providing facilities, infrastructures, and enterprises**







Fellows are **hosted by EMBL group and team leaders** across all sites and perform one or more **secondments with more than 50 partner organisations** around the globe

EMBL-EBI

Industry engagement



www.ebi.ac.uk/**industry**

EMBL-EBI

Supporting industry

- Our Industry Programme offers neutral ground where companies can meet and explore new developments and shared challenges.
- We support small and medium enterprises by providing free data, tools and infrastructure

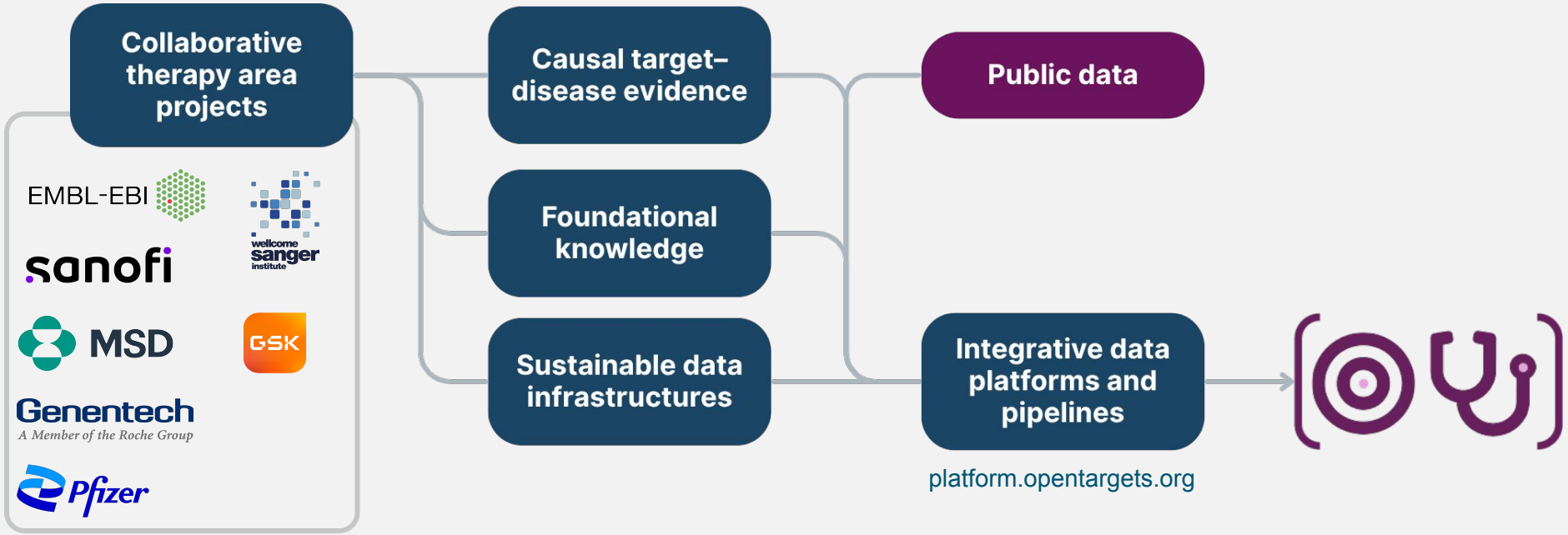


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Open Targets

Open Targets

Pre-competitive insights for systematic drug target selection | **opentargets.org**





ELIXIR

Connecting, coordinating and integrating Europe's life science data resources

EMBL-EBI as an ELIXIR Node

www.elixir-europe.org

elixir

EMBL-EBI

ELIXIR

ELIXIR is a European research infrastructure which manages and safeguards the massive amounts of life science data and tools being generated by publicly-funded research.

We are a network of scientific and technical experts working together to coordinate, integrate and sustain bioinformatics resources, providing access to high-quality data, tools, standards, compute and training.

These open resources enable researchers to access and analyse life science data to accelerate research and its impact on society.



EMBL-EBI

ELIXIR members

ELIXIR members host Nodes, including EMBL-EBI, which represent centres of excellence in bioinformatics.

This includes more than 850 people from over 245 research institutes.

ELIXIR activities are coordinated by the ELIXIR Hub, based at the Wellcome Genome Campus, UK.



EMBL-EBI

ELIXIR – what do we do

We build life science informatics **capacity and infrastructure** in Europe, connect and develop a **network of experts** and provide hundreds of high quality **services and resources** available to all

- Databases
- Training
- Software tools
- Data standards
- Compute resources
- Scientific & technical experts



EMBL-EBI

European Nucleotide Archive



EMBL-EBI

ENA History

1980

EMBL Data Library, Los Alamos National Laboratory and DNA Databank of Japan collaborate

First mandated submissions

Collaboration with International Patent Offices

Expansion to cover next generation platforms

Engagement with data standards bodies

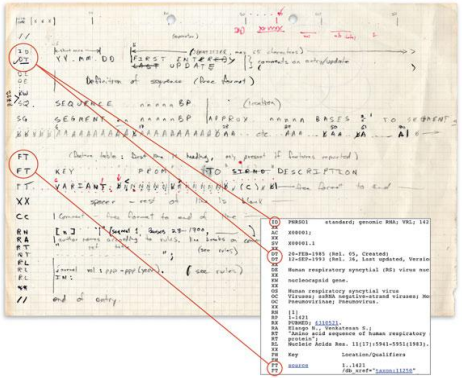
Data submissions brokering

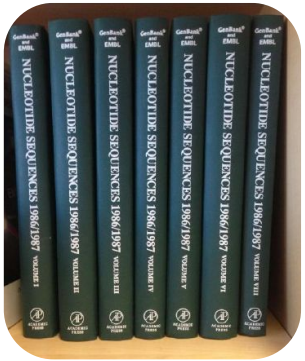
Engagement with UN processes

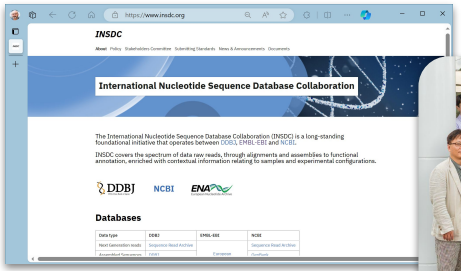
Minimal Specifications

Global Participation Programme

2026









- DNA Data Bank of Japan (DDBJ) at the National Institute of Genetics in Mishima, Japan

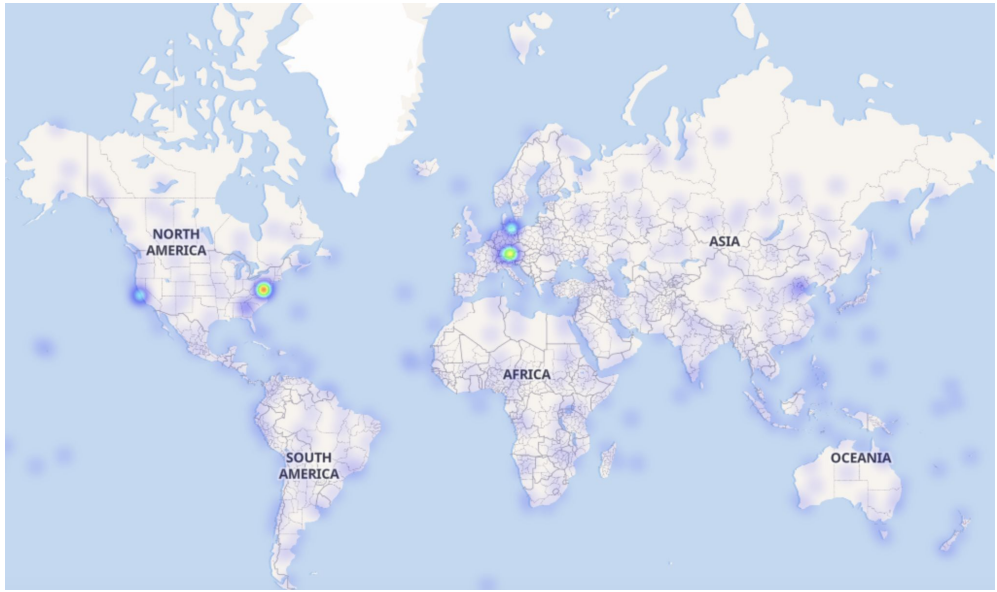
- European Nucleotide Archive (ENA) at the European Molecular Biology Laboratory's European Bioinformatics Institute (EMBL-EBI) in Hinxton, UK

- GenBank at National Center for Biotechnology Information (NCBI), National Library of Medicine, National Institutes of Health in Bethesda, Maryland, USA

Karsch-Mizrachi I, Arita M, Burdett T, Cochrane G, Nakamura Y, Pruitt KD, Schneider VA, Of The International Nucleotide Sequence Database Collaboration OB. **The international nucleotide sequence database collaboration (INSDC): enhancing global participation.** Nucleic Acids Res. 2025 Jan 6;53(D1):D62-D66. doi: 10.1093/nar/gkae1058.

Scale

- >90 petabytes of core data (before replication)
- >100 countries
- >24,000 submission transactions per day; 1 every 4 seconds



EMBL-EBI

Data submissions and brokering



Submissions

Submissions over the last year include 5.5 million runs, 4.4 million samples and 1.9 million assemblies

More than 2800 data submitters from 88 countries submitted



Data brokering network

62 brokers

19 countries represented

Brokered submissions account for 13% of raw reads and 21% assemblies in ENA

Priority broker helpdesk queue



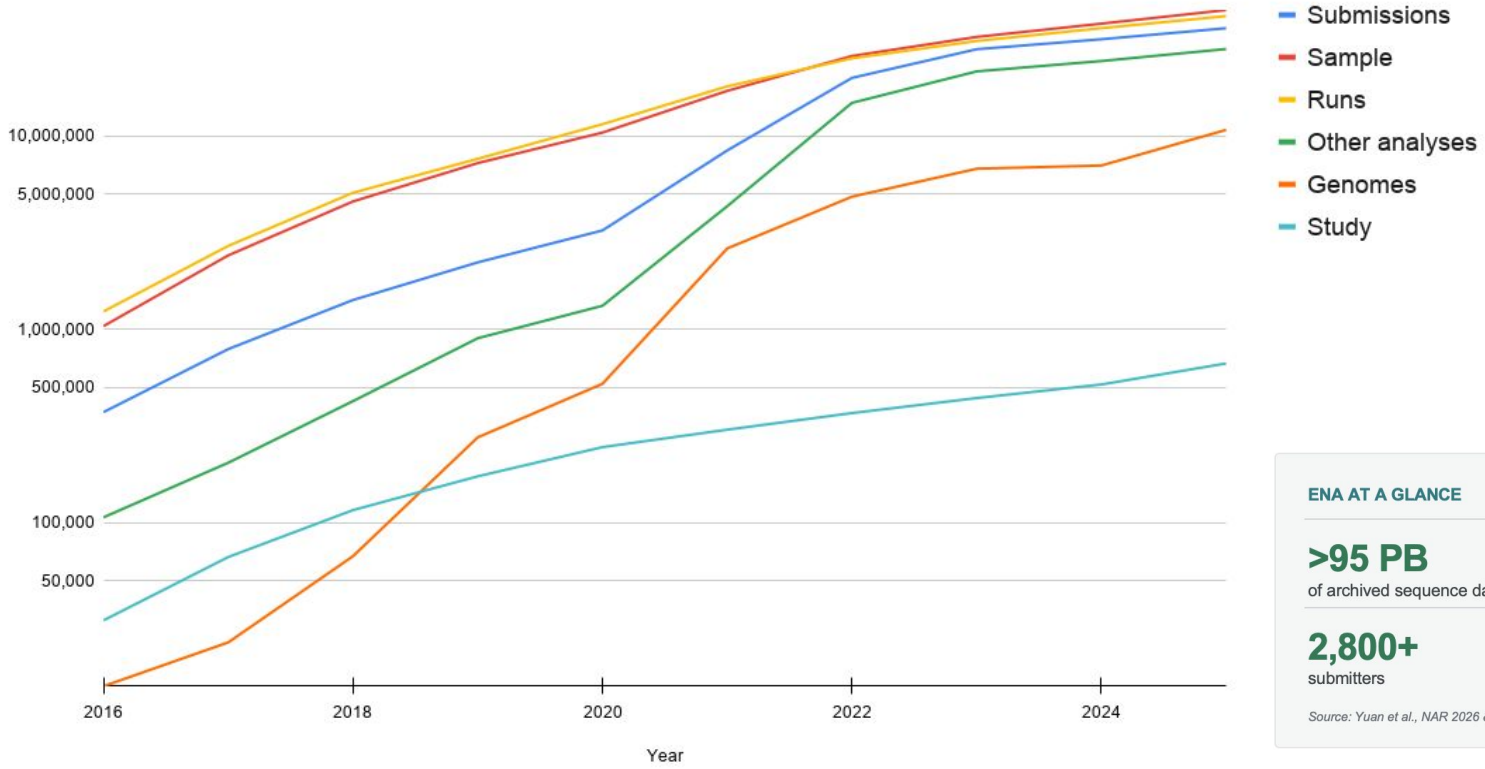
Worldwide use

Serving users from 200 countries

500k visitors/month

192 million API requests/month

ENA Cumulative Record Count



ENA AT A GLANCE

>95 PB
of archived sequence data

~500k
monthly users

2,800+
submitters

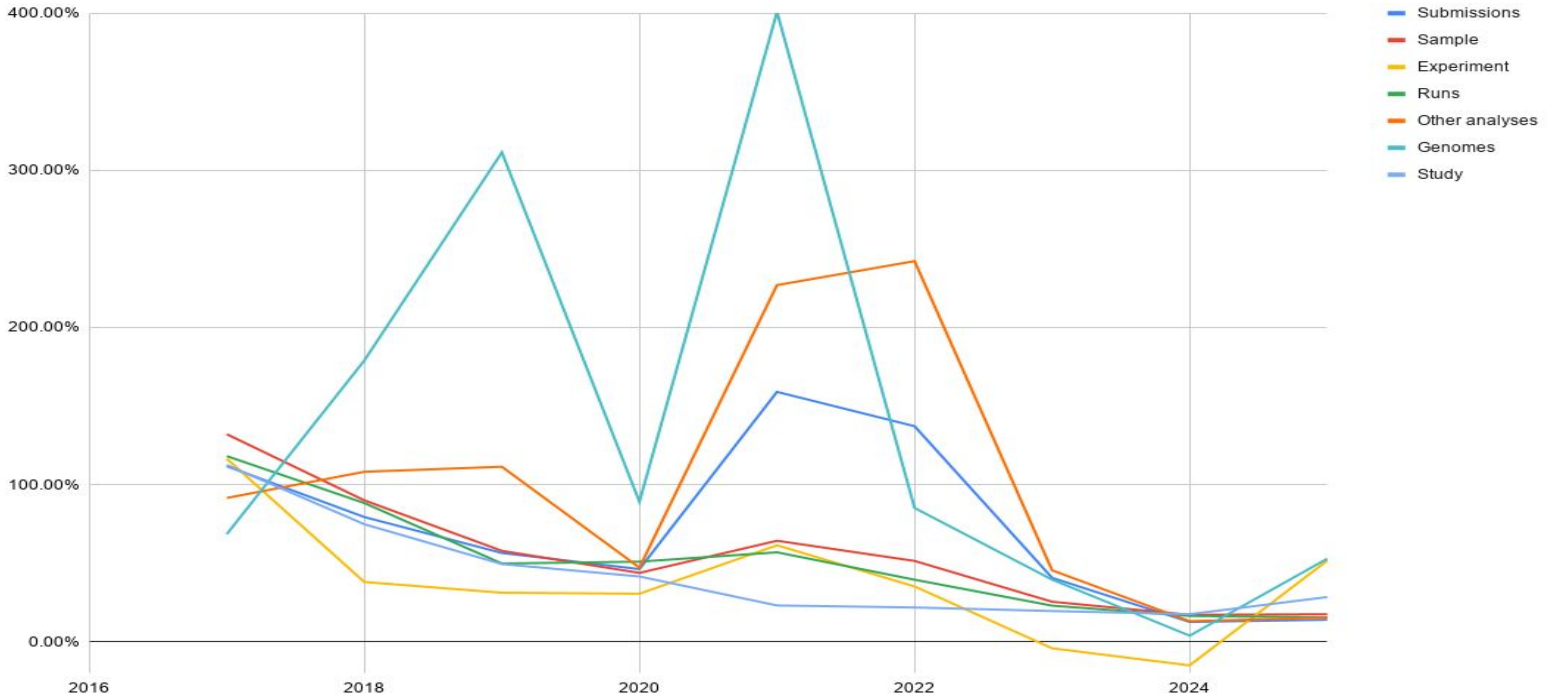
88
countries represented

Source: Yuan et al., NAR 2026 & Internal metrics

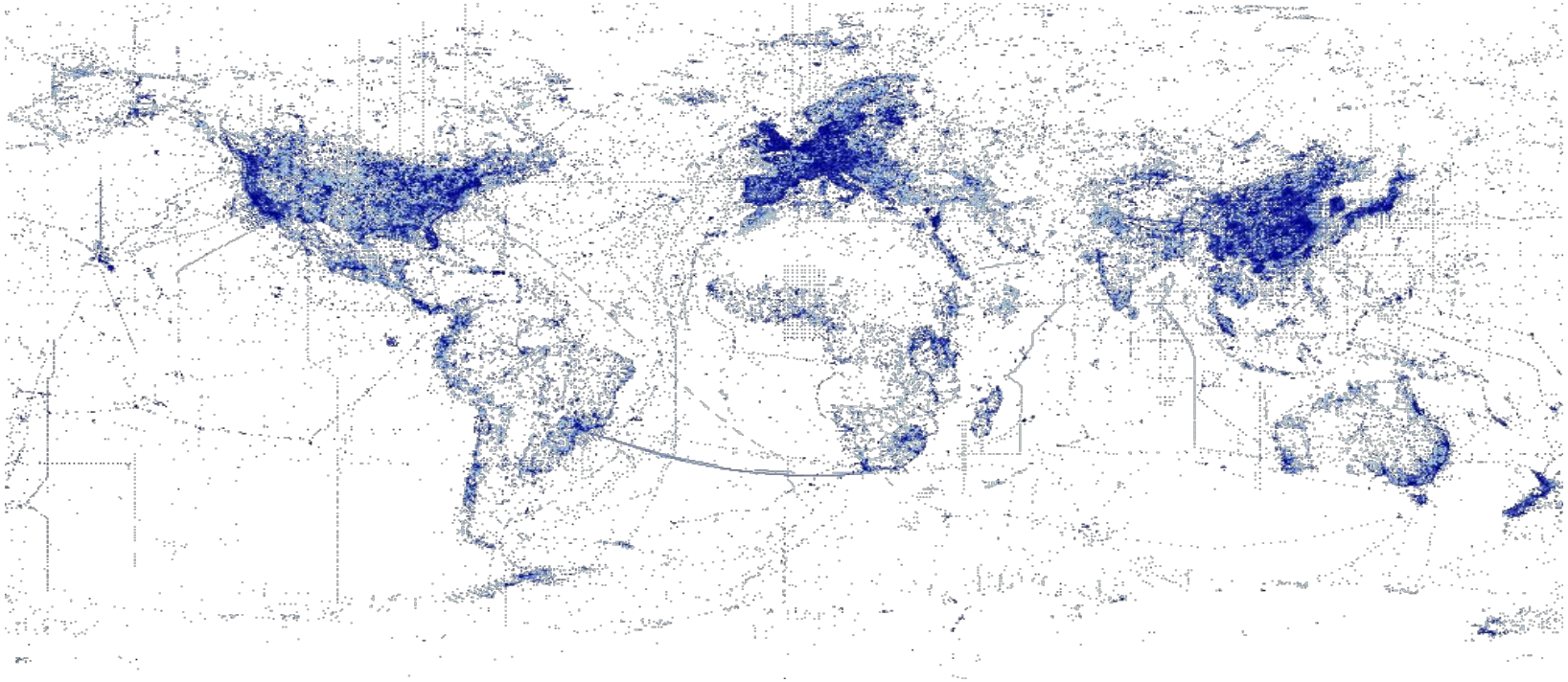
ENA NAR 2025: https://doi.org/10.1093/nar/gkaf1295

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ENA Growth rate



Where does ENA data come from?



EMBL-EBI

FAIR Data and support



Data standards

New checklist editor

JSON-based

Support for versioning of checklists to ensure of backward compatibility

Continuing work with standards initiatives and organisations



Support

Helpdesk supporting an average of 310 user queries a month

Reducing helpdesk response times

Exploring AI/ML applications for helpdesk support with EMBL-EBI colleagues



Communications

Launch of BlueSky account

Outreach/training at 30 events.

Metadata standards

- Continued engagement with standards communities: GSC, TDWG, GA4GH
- Three new sample checklists available
  - Ancient DNA (ERC000059)
  - ENA marker sequence (ERC000060)
  - Laboratory animal (ERC000061)
- 62 new terms
- Framework for experiment metadata

genomic STANDARDS consortium

minas

Biodiversity Information Standards TDWG

Global Alliance for Genomics & Health

Make eDNA FAIR

Findable
Accessible
Interoperable
Reusable

AEGIS
Ancient Environmental Genomics Initiative for Sustainability

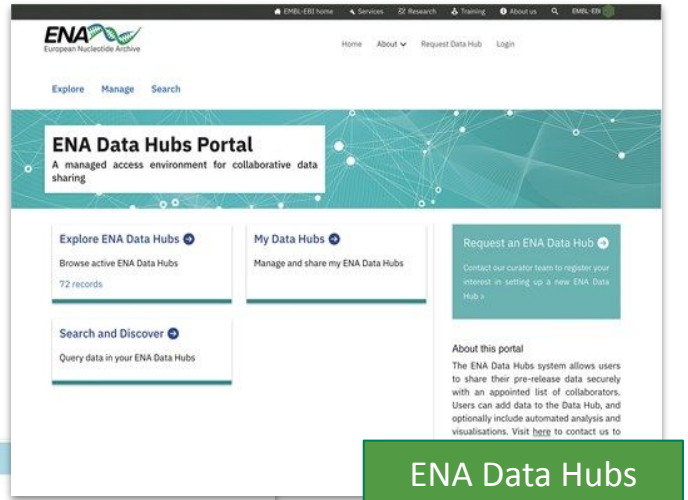
Biodiversity Genomics Europe

eDNAqua-Plan
NEXT GENERATION OF AQUATIC BIODIVERSITY MONITORING

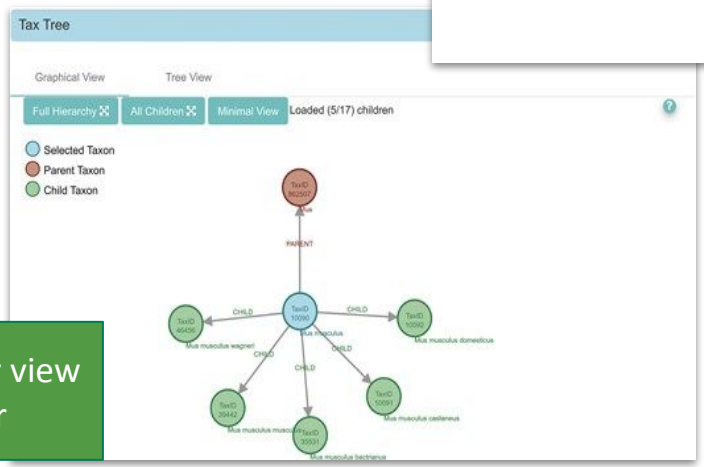
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Service enhancement updates

- Deployment of GraphDB to improve visualisations of Taxonomy tree
- ENA Data Hubs Portal, expanding the Data Hubs beyond infectious disease and automating setup and management
- ENA / Biosamples integration



ENA Data Hubs Portal

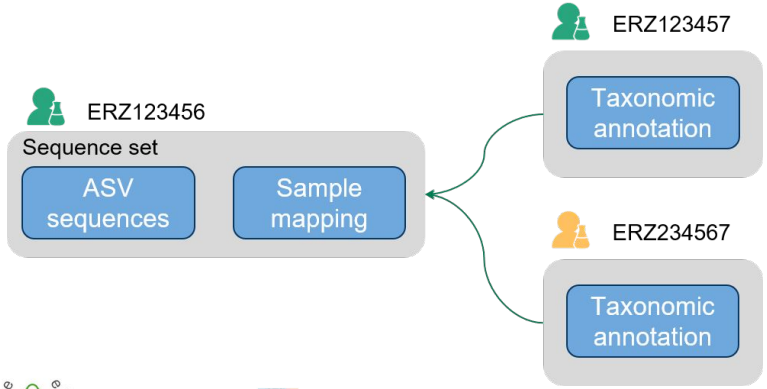


Graphical taxonomy view in ENA Browser

EMBL-EBI

eDNA object updates

- **Environmental sequence set** object implemented for submission of Amplicon Sequence Variants (ASVs) derived from eDNA single locus amplification from multiple samples
- Decoupled submission of sequences/ASV table and taxonomic annotation
- Rich metadata following community standards
- Testing phase with external partners



Sequence_id	Tax_id	scientific_name	sample_1	sample_2	sample_3	
OTU1	70284	Microtus cabrerae	29	23	17	CTAAGTTTCTCGAGCTTTCTCTTCCAGAATTCAAAGA
OTU2	10129	Apodemus sylvaticus	10	475	5	ATCCCAAGGAATATTTTTATTCCAGCCCTGGGAAATA
OTU3	10096	Mus spretus	8	12	3	CAAGGAATATTTTTATTCCAGCCCTGGGAAATATTAC
OTU4	36802	Crocidura russula	11	1	1	AGCTTTCTCTTCCAGAATTCAAAGAATTGAGTATCCC
OTU5	184258	Microtus lusitanicus	4	5	3563	CTAAGTTTCTCGAGCTTTCTCTTCCAGAATTCAAAGA
OTU6	88454	Arvicola sapidus	0	0	0	ATCCCAAGGAATATTTTTATTCCAGCCSTGGGAAATA

Processed Sequences

```
>AGRIVOLE_ASV1
ACTATGCTTAGCCTTAAACTTAAATAATTTAATAACAAAATTATTTGCCAGAGAGCTACTAGCTACTGCTTAAAACTCAAA
>AGRIVOLE_ASV2
GCTATGCTTAGCCATAAACCTAAATAATTAAATTTAACAAAACTATTTGCCAGAGAACTACTAGCCATAGCTTAAAACTCA
>AGRIVOLE_ASV3
ACTATGCTTAGCCTTAAACTTAAATAATTTAATAACAAAATTATTTGCCAGAGAACTACTAGCTACTGCTTAAAACTCGAA
>AGRIVOLE_ASV4
ACTATGCTTAGCCCTAAACCTCAACAGTTAAATCAACAAAACTGCTCGCCAGAACACTACGAGCCACAGCTTAAAACTCAA
>AGRIVOLE_ASV5
ACTATGCTTAGCCCTAAACCTCAATAATTTAGAAACAAAAATATTTGCCTGAGAACTACTGGCCACAGCTTAAAACTCAAA
>AGRIVOLE_ASV6
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>AGRIVOLE_ASV7
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>AGRIVOLE_ASV8
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>AGRIVOLE_ASV13
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>AGRIVOLE_ASV14
ACTATGCTTAGCCCTAAACTAAAACAATCAAATAACAAGACTGTTCGCCAGAGAACTACTAGCAATAGCTTAAAACTCAAA
```

OTU/ASV table / sample map

Sequence_id	Sample_id	Frequency
AGRIVOLE_ASV1	SAMEA11243268	14
AGRIVOLE_ASV2	SAMEA11243267	13
AGRIVOLE_ASV3	SAMEA11243268	18
AGRIVOLE_ASV4	SAMEA11243259	4
AGRIVOLE_ASV4	SAMEA11243264	97
AGRIVOLE_ASV4	SAMEA11243266	4
AGRIVOLE_ASV4	SAMEA11243267	5
AGRIVOLE_ASV4	SAMEA11243268	5
AGRIVOLE_ASV4	SAMEA11243270	1
AGRIVOLE_ASV5	SAMEA11243259	8
AGRIVOLE_ASV5	SAMEA11243261	8
AGRIVOLE_ASV5	SAMEA11243265	3
AGRIVOLE_ASV6	SAMEA11243260	30
AGRIVOLE_ASV6	SAMEA11243270	8
AGRIVOLE_ASV7	SAMEA11243268	24
AGRIVOLE_ASV8	SAMEA11243259	2
AGRIVOLE_ASV8	SAMEA11243261	5
AGRIVOLE_ASV8	SAMEA11243265	8
AGRIVOLE_ASV8	SAMEA11243266	12
AGRIVOLE_ASV8	SAMEA11243268	1
AGRIVOLE_ASV9	SAMEA11243268	14
AGRIVOLE_ASV10	SAMEA11243259	5
AGRIVOLE_ASV10	SAMEA11243265	6

Taxonomic annotation

Sequence_id	Tax_id
AGRIVOLE_ASV1	9031
AGRIVOLE_ASV2	9606
AGRIVOLE_ASV3	1949063
AGRIVOLE_ASV4	88454
AGRIVOLE_ASV5	3119508
AGRIVOLE_ASV6	10129
AGRIVOLE_ASV7	9606
AGRIVOLE_ASV8	10096
AGRIVOLE_ASV9	2836371
AGRIVOLE_ASV10	88454
AGRIVOLE_ASV11	10129
AGRIVOLE_ASV12	2922388
AGRIVOLE_ASV13	70284
AGRIVOLE_ASV14	10129
AGRIVOLE_ASV15	9606
AGRIVOLE_ASV16	69158
AGRIVOLE_ASV17	9606
AGRIVOLE_ASV18	70284
AGRIVOLE_ASV19	10096
AGRIVOLE_ASV20	2922388
AGRIVOLE_ASV21	9606
AGRIVOLE_ASV22	32644

Make eDNA FAIR — Findable Accessible Interoperable Reusable

Biodiversity Genomics Europe

BiCIKL

eDNAqua-Plan — NEXT GENERATION OF AQUATIC BIODIVERSITY MONITORING

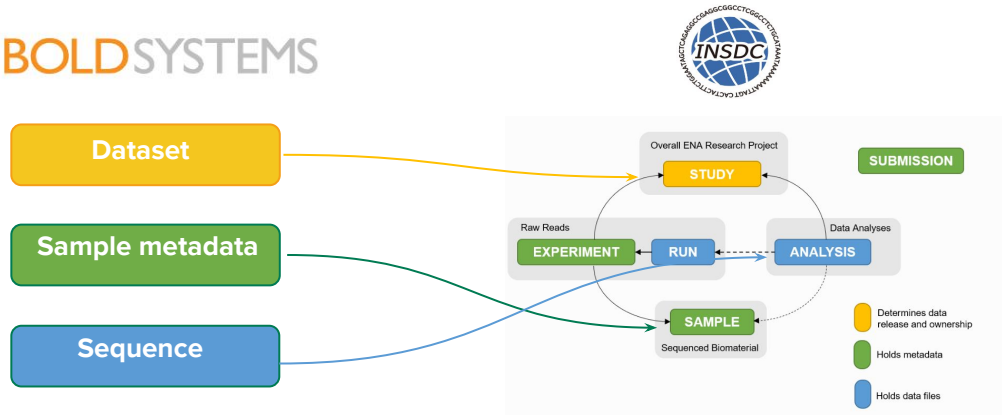
EMBL-EBI

Improving attribution and ABS

- EMBL-EBI response to Decision 16/2 of the Convention on Biological Diversity
  - A checkbox prompt for data submitters asking them to confirm that the data they are submitting is not subject to any ABS restrictions
  - Updates to public facing documentation and pages (eg submission policies and procedures)
  - A mechanism for linking to sample spatiotemporal minimal metadata
- Working towards implementation by Oct 2026 (next CBD COP meeting)

BOLD data exchange updates

- BOLD is setting up a data submission pipeline to ENA - for European data – beta testing
- BOLD setting up sample submission of the specimen metadata as well as the sequence submission
- ENA implemented the ENA Marker Sequence checklist (based on MIxS MIMARKS and some additional optional metadata)

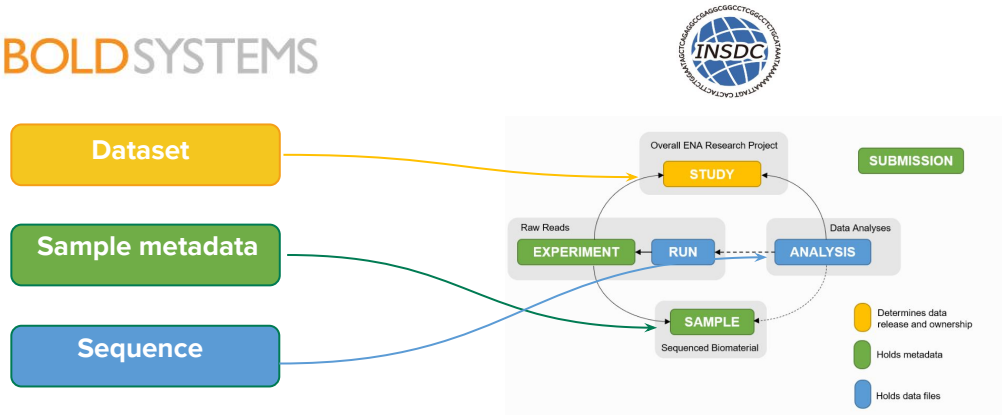


Data model integration

EMBL-EBI

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Data model integration

EMBL-EBI

Ongoing and Future direction

Data Model	Infrastructure	Discovery	User Experience
Completing decoupled annotation system	BioSamples platform move to Kubernetes	Indexing ENA data for search incrementally	ENA Browser revamp
EMBL Flatfile ⟶ GFF3 (Independent annotation layer)	Migration from SQL database to noSQL databases	Indexing BioSamples data with Solr	Improved helpdesk queue monitoring
Continued standards and community engagement	Minimising presence of certain read format		

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